Banking System API Specification

Version: 1.0.0

Base URL: https://nodejs-based-banking-app-backend.onrender.com

GitHub Repo URL: https://github.com/aryanm9026/nodeJS-based-banking-app-backend

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Overview

This API service provides a secure, ACID-compliant backend for banking operations. It features an immutable ledger, JWT-based authentication, and idempotency protection for all financial transactions.

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1. Authentication

Endpoint	Method	Description	Payloads
/api/auth/register	**POST**	Registers a new user. Required fields: name, email, password.	"name": "your_name", "email": "your_email", "password": "your_password"
/api/auth/login	**POST**	Authenticates user and returns a 3-day JWT.	"email": "your_email", "password": "your_password"
/api/auth/logout	**POST**	Invalidates the current token via blacklisting.	-

2. Account Management

Requires Header: Authorization: Bearer <token>

Endpoint	Method	Description	Payloads
/api/accounts/	**POST**	Creates a new ACTIVE account for the user.	-
/api/accounts/	**GET**	Retrieves all accounts owned by the user.	-

Endpoint	Method	Description	Payloads
/api/accounts/balance/:id	**GET**	Derives real-time balance from the ledger.	Requires the 'accountId' as a URL parameter

3. Transaction Engine

Requires Header: Authorization: Bearer <token>

Endpoint	Method	Description	Payloads
/api/transactions/	**POST**	Executes a fund transfer. Required: fromAccount, toAccount, amount, idempotencyKey.	"fromAccount": "67f1a...", "toAccount": "67f2b...", "amount": 1000, "idempotencyKey": "unique-uuid-v7-string"
/api/transactions/system/initial-funds	**POST**	Restricted route for System Users to inject liquidity. (System Users Only Route)	"toAccount": "67f2b...", "amount": 5000, "idempotencyKey": "system-fund-unique-key"

4. Error Code You May See

Code	Meaning	Context
400	**Bad Request**	**Missing fields, insufficient balance, or account inactive.**
401	**Unauthorized**	**Missing, invalid, or blacklisted token.**
403	**Forbidden**	**Attempting to access "systemUser" routes without permission.**
404	**Not Found**	**Specified account or user does not exist.**
422	**Unprocessable Entity**	**Registration with an email that is already in use.**
500	**Internal Error**	**Database connection failure or unexpected server crash.**